

TrES-3b

R-2699

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_210616 · created 2026-09-01T00:09:11+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459381.9229 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.015
Transit depth (Rp/Rs)^2	3.14 +/- 0.54 [%]
Orbital Inclination (inc)	84.47 +/- 0.93
Airmass coefficient 1 (a1)	0.953 +/- 0.029
Airmass coefficient 2 (a2)	0.04 +/- 0.025
Scatter in the residuals of the lightcurve fit is	3.05 %
Best Comparison Star	#3 - [317, 291]
Optimal Aperture	3.31
Optimal Annulus	7.26
Transit Duration (day)	0.0732 +/- 0.0048

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.177 $\pm$ 0.015 vs 0.16309 (deviation 0.63 σ)
Duration fit vs published	0.0732 d vs 0.0567 d (deviation 2.36 σ)
Mid-time O–C	3.1 min
Depth SNR	8.1
Residual scatter	3.05 %

Light curve

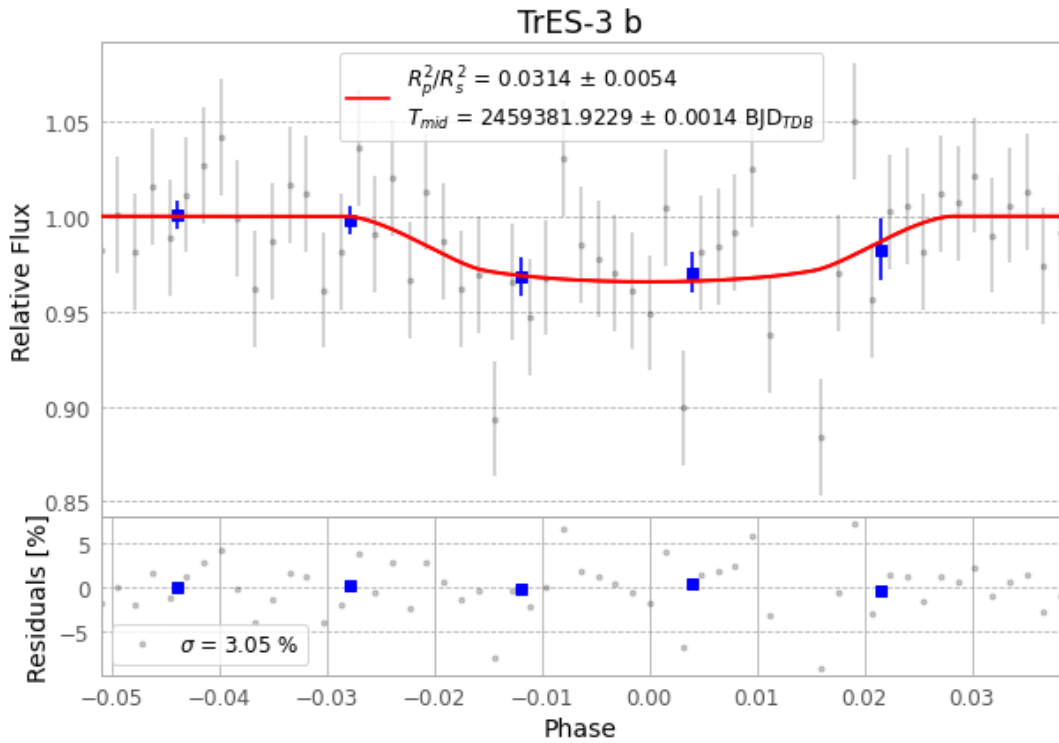


Figure 1 — FinalLightCurve_TrES-3 b_2021-06-16.png (SHA-256 ea0f114eea8f0146...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [388, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fb4a37e1141d7a0a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

57 FITS frames (37 MB), TRES-3210616082712.FITS → TRES-3210616111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-210616.log (09c820e0964c...), FinalLightCurve_TrES-3 b_2021-06-16.png (ea0f114eea8f...), FinalLightCurve_TrES-3 b_2021-06-16.csv (635ea14de3df...)

Compute resources

Wall time 107.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:09Z.